

System Project - DREAM Car Setup & Development Reference

AI for Robotics — DREAM Car Setup & Development Guide

Set up your connection, write your policy, deploy it to the car, and test it safely

This guide takes you through the complete workflow for working with your team's **DREAM car**.

You will write and edit your policy on **your own computer**, copy it to the car when you are ready to test, and use **Foxglove** to control and monitor the vehicle.

Your normal development loop

Edit locally → Copy to the car → Build → Run → Observe → Improve

New to the cars?

Start with **1. SSH and VPN Setup** and work through the numbered pages in order.

Core workflow

These are the main pages you will use when working with the car.

Step 1

SSH and VPN Setup

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-1-ssh-and-vpn-setup>)

Connect your computer to the University network, find your car's IP address, and log in over SSH.

[Open SSH and VPN Setup →](#)

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-1-ssh-and-vpn-setup>).

Step 2

Clone and Edit Your Policy

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-2-clone-and-edit-your-policy>)

Clone `ai4r_policy` onto your own computer, understand the policy interface, and write your code in `policy_node.py`.

[Open Clone and Edit Your Policy →](#)

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-2-clone-and-edit-your-policy>).

Step 3

Put Your Code on the Car

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-3-transfer-and-build>)

Transfer your policy with `scp`, build the student workspace, and prepare the new version for testing.

[Open Transfer and Build →](#)

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-3-transfer-and-build>).

Step 4

Run the Car

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-4-run-the-car>)

Start the required runtime units, enable the vehicle, run your policy, stop safely, and inspect runtime logs.

[Open Run the Car →](#)

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-4-run-the-car>).

Step 5

Foxglove: Driving the Car and Watching It

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-5-foxglove-ui>)

Connect the Foxglove interface, use the vehicle and policy controls, and inspect live sensor data and commands.

[Open Foxglove Guide →](#)

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-5-foxglove-ui>).

Once you are set up

You do not need to repeat every setup step whenever you change your policy. During a normal development session, your workflow is:

1. Edit your policy on your own computer

2. Copy the changed file to the car with `scp`

3. Run `dream build ros student`

4. Restart `ai4r_policy`

5. Run the policy again and inspect the result in Foxglove

Remember:

the copy on your own computer is your source of truth. Do not rely on the car to store the current version of your work between sessions.

Companion guides

These pages cover common situations that are useful once you have the basic workflow running.

Policy development

Pass Your Notebook Policy to the Car

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-2a-pass-your-notebook-policy-to-the-car>)

Translate a policy from the Dream Gym notebook to the real vehicle. Covers observation names, cone colour IDs, action outputs, LiDAR differences, timing, and common porting mistakes.

[Open Notebook → Car Guide →](#)

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-2a-pass-your-notebook-policy-to-the-car>)

Team collaboration

Working as a Team with Git

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-2b-working-as-a-team-with-git>)

Set up a private team fork, add teammates, use branches for different ideas, and keep your fork up to date with framework changes.

[Open Team Git Workflow →](#)

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-2b-working-as-a-team-with-git>)

Optional convenience

Set Up an SSH Key

(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-optional-setup-an-ssh-key>)

Configure password-free SSH and `scp`. This is optional — using the workshop password is always sufficient.

[Open SSH Key Setup →](#)

<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-optional-setup-an-ssh-key>

Which page do I need?

I want to...	Go to
Connect to my car for the first time	<u>1. SSH and VPN Setup</u> <u>https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-1-ssh-and-vpn-setup</u>
Find where to write my policy	<u>2. Clone and Edit Your Policy</u> <u>https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-2-clone-and-edit-your-policy</u>
Move my notebook policy onto the real car	<u>Notebook → Car Guide</u> <u>https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-2a-pass-your-notebook-policy-to-the-car</u>
Copy my latest changes to the car	<u>3. Put Your Code on the Car</u> <u>https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-3-transfer-and-build</u>
Start, stop or restart my policy	<u>4. Run the Car</u> <u>https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-4-run-the-car</u>
See cones, sensor data or vehicle states	<u>5. Foxglove</u> <u>https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-5-foxglove-ui</u>
Work on the policy with several teammates	<u>Working as a Team with Git</u> <u>https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-2b-working-as-a-team-with-git</u>

I want to...	Go to
Stop typing the SSH password every time	Optional SSH Key Setup (https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-optional-setup-an-ssh-key)

Working with a physical car

Keep the RC remote within reach while testing, never leave a running car unattended, and do not enable another team's vehicle.

Follow the start and stop sequence in [4. Run the Car](https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-4-run-the-car)
(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-4-run-the-car>).

Ready to start?

Begin with [1. SSH and VPN Setup](https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-1-ssh-and-vpn-setup) →
(<https://canvas.lms.unimelb.edu.au/courses/238681/pages/system-project-docs-1-ssh-and-vpn-setup>).